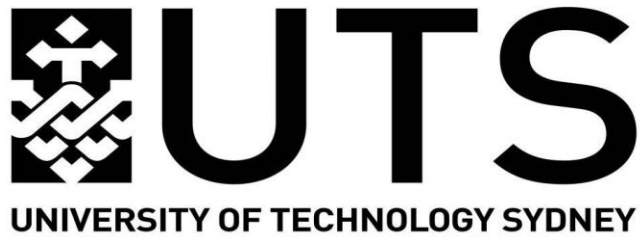


UNIVERSITY OF TECHNOLOGY SYDNEY



42048 Studio 3: 22789 Financial Analytics - Spring 2025

Assessment task 2: Balanced Scorecard Case Study

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Table of Contents

Part A: Financial Analysis Dashboard	4
Part B: Balanced Scorecard Development	6
1. Customer satisfaction	6
2. Internal process efficiency	7
3. Employee learning and growth	9
4. Social performance	10
5. Environmental sustainability	11
6. Indigenous data sovereignty principles and ethical data collection and reporting practices	13
Reference	14
Appendix — Governance Checklist	14

Table of Figure

Figure 1 Overview of the company	4
Figure 2 OPM and GPM analysis Nov2024	5
Figure 3 Customer Satisfaction KPIs	6
Figure 4 Internal Process Efficiency KPIs.....	8
Figure 5 Employee Learning And Growth KPIs	9
Figure 6 Social Performance KPIs.....	10
Figure 7 Environmental Sustainability KPIs	12

Financial Dashboard Analysis Report - Jubilee Superstores Ltd

Part A: Financial Analysis Dashboard

Overview page: design rationale and key signals

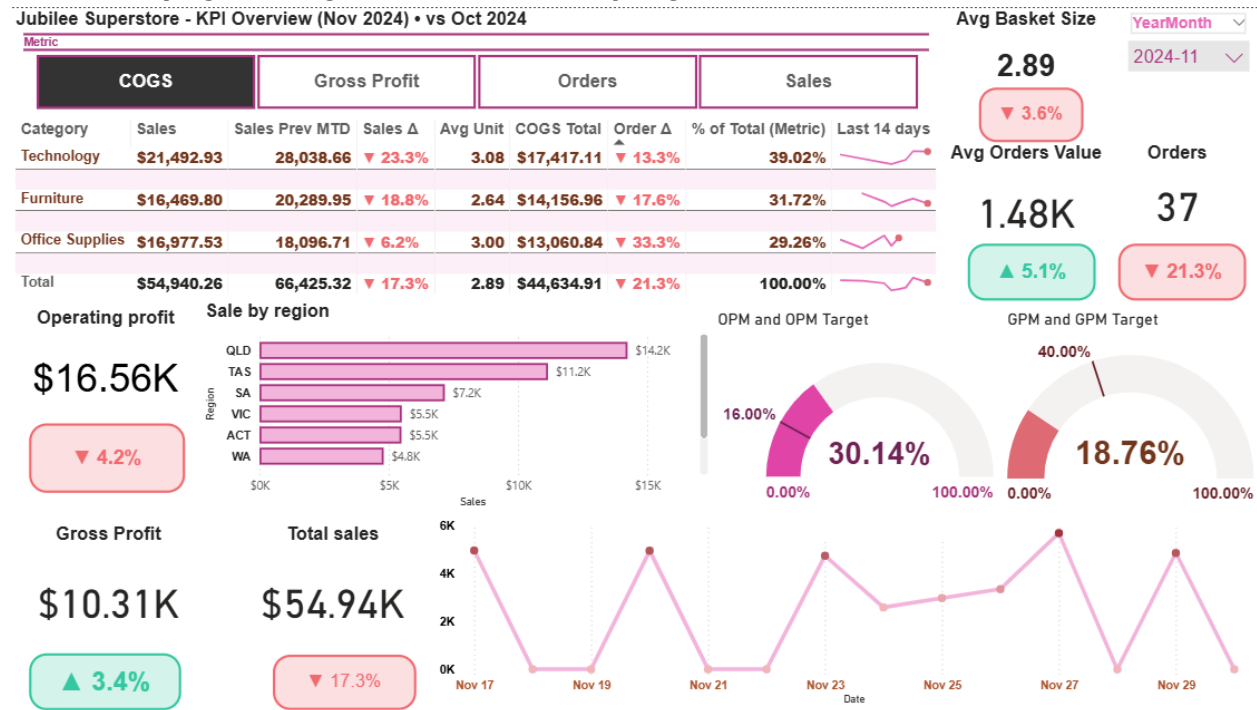


Figure 1 Overview of the company

The dashboard answers “How did Jubilee perform in Nov-2024?” using a YearMonth slicer fixed to Nov-2024 and a metric switch (COGS, Gross Profit, Orders, Sales) so views stay comparable. Headline KPIs show revenue ≈ \$54.9K, gross profit \$10.3K, 37 orders, basket size 2.89. Margin cards surface the signal: OPM 30.14% (above target) vs GPM 18.76% (well below the 40% target), so the issue is gross-margin creation, not operating cost. Category/region bars combine share-of-total with current MTD; 14-day sparklines show momentum. Product bullet bars plot current vs previous with constant target lines to make gaps obvious. Two ranked tables convert variance to dollars (OP Excess \$, GP Shortfall \$) to focus action. Together, the page communicates performance, trend, and where to intervene.

OPM & GPM analysis:

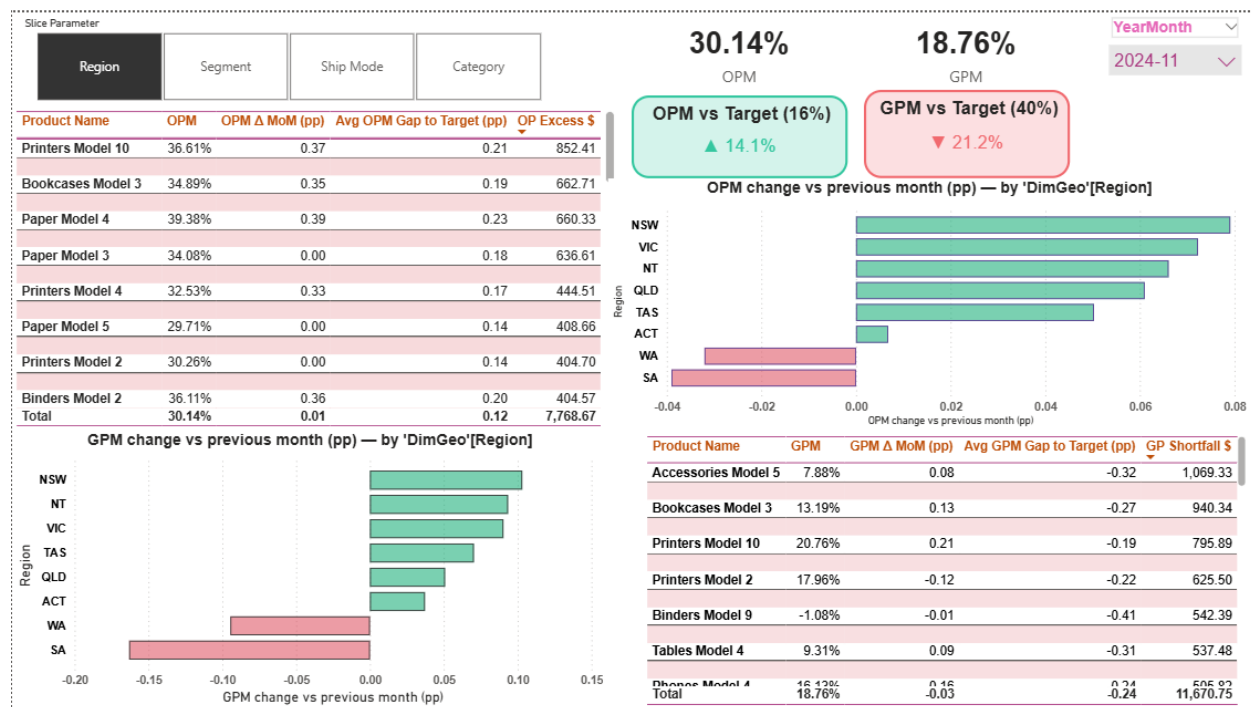


Figure 2 OPM and GPM analysis Nov2024

Because OPM and GPM answer different economic questions, I separated their explanations while keeping the visual grammar identical to reduce cognitive load. For each metric, the product bars with target lines expose how individual SKUs contribute to the company-level result; the accompanying MoM change (pp) small-multiple shows whether the trajectory is improving or weakening across the selected slice. The MoM bars are centred on zero so direction and magnitude are immediately apparent; a tooltip reports Current, Previous, and Δ (pp) for auditability. This layout was chosen over cumulative or stacked forms because managers need to isolate movement and distance-to-target, not just totals. Expressing change in points rather than relative percentages ensures a one-to-one interpretation: a +0.06 pp improvement in NSW OPM is the same unit everywhere in the business.

Interpreting Nov-2024 through this lens, the company converts operating costs efficiently- OPM at 30.14% is roughly +14.1 pp above the 16% threshold and improving in most eastern regions. In contrast, the gross margin engine is underperforming- GPM at 18.76% is about –21.2 pp below the 40% KPI, and although some regions show small positive MoM moves, several markets (notably ACT and WA) remain negative. The GP Shortfall \$ table identifies where that gap costs the most in absolute dollars-Accessories

Model 5, Bookcases Model 3, Printers Models 10 and 2, Binders Model 9, and Tables Model 4- which provides a surgical starting list for remediation. The OPM Excess \$ view highlights SKUs such as Printers Model 10 and Paper Model 4 that are consistently outperforming; these should be protected from stock-outs and service erosion so the operating leverage is preserved.

The practical implication is to run a two-track intervention. First, address GPM directly through price and discount governance, supplier cost conversations, and mix management concentrated on the highest dollar shortfalls and the regions with persistent negative MoM readings. Second, maintain OPM discipline where declines appear (such as SA/WA) by reviewing labour scheduling and freight utilisation, but avoid blunt cost cuts that could further depress gross margin. Re-running the dashboard each month with the same parameterised views will make improvement measurable: the MoM bars should move rightwards, the shortfall table should shrink in dollars, and the overview KPI tiles should flip to green for both OPM and GPM.

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Part B: Balanced Scorecard Development

1. Customer satisfaction

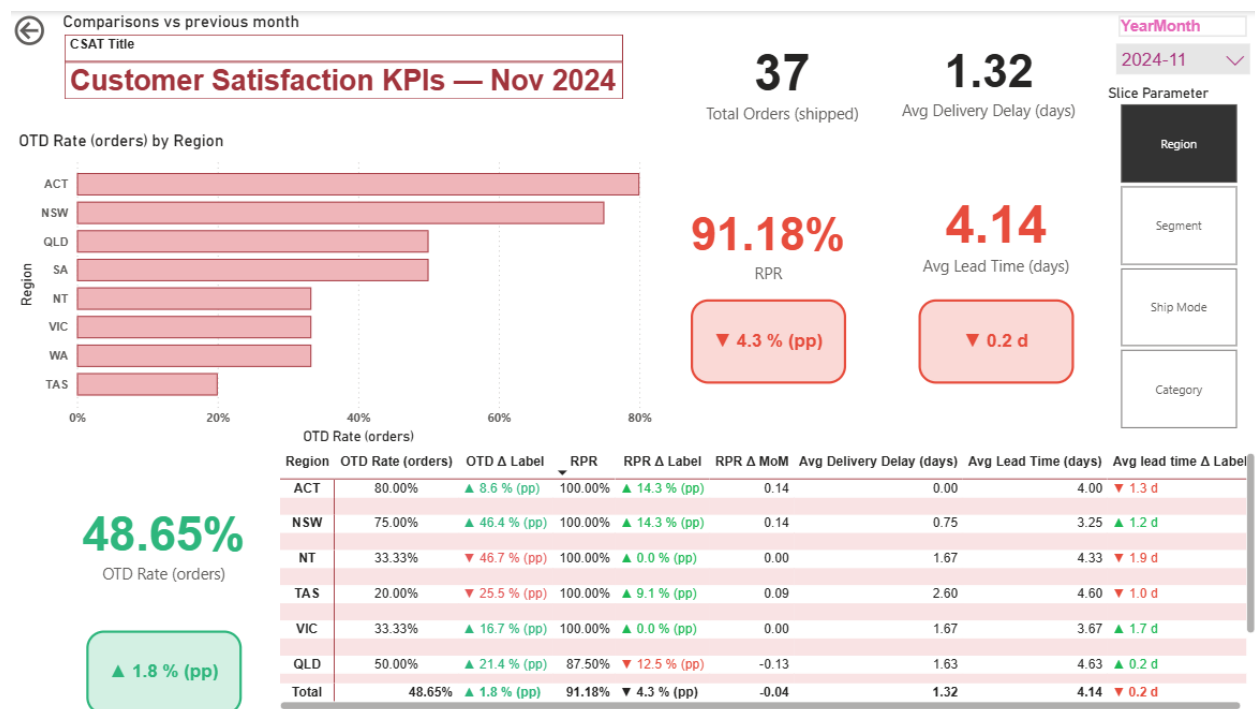


Figure 3 Customer Satisfaction KPIs

KPI 1- On-Time Delivery (OTD) rate.

I select OTD as the primary customer-satisfaction KPI because it directly reflects service reliability - the first thing customers judge. OTD is calculated from the Orders table by mapping an SLA to each Ship Mode (Same-day=0, First=2, Second=4, Standard=6 days), deriving Promised Date = Order Date + SLA, flagging OnTime = Ship Date $\leq$ Promised Date, and dividing on-time shipments by total shipments in the month. In Nov-2024 our OTD is 48.65% (rise 1.8 pp MoM) with wide regional disparity: ACT $\approx$ 80%, NSW $\approx$ 75%, while WA/TAS/VIC/NT $\leq$ 33%. Although Average Lead Time eased to 4.14 days (-0.2 d), Average Delivery Delay remains 1.32 days, indicating execution issues rather than customer promise setting. Improving OTD supports the strategic goal of customer-centric growth by reducing complaints, refunds and churn. The KPI links to other dimensions used on the page- Region, Segment, Category, Ship Mode- to localise root causes and prioritise lanes and products.

KPI 2 - Repeat-Purchase Ratio (RPR).

RPR measures loyalty and Lifetime Value. Using Customers and Orders, I count customers with ≥ 2 orders in the period and divide by all active customers; MoM is tracked with a date dimension. In Nov-2024 RPR = 91.18% (-4.3 pp MoM). Regions WA (0.67), SA (0.80) and QLD (0.88) lag others (often 1.00), consistent with poor OTD, linking experience to loyalty. Required fields are Customer ID, Order Date, and Region/Segment/Category to attribute repeat behaviour. Strategically, RPR aligns to profitable retention - repeat buyers lower acquisition cost and stabilise revenue mix-and connects back to financial KPIs (e.g., better RPR supports higher OPM). Together, OTD and RPR provide a clear line of sight from operations to loyalty; the dashboard's slicers allow managers to act where reliability is weakest and loyalty at risk.

2. Internal process efficiency

KPI- Internal Process Efficiency

Category	COGS Total	CPU (COGS per Unit)	CESR	CPU Δ	CESR Δ
Technology	\$17,417.11	435.43	46.15%	▼ 62.2	▼ 0.51%
Furniture	\$14,156.96	382.62	57.14%	▼ 19.2	▲ 15.97%
Office Supplies	\$13,060.84	435.36	50.00%	▲ 137.6	▼ 3.33%
Total	\$44,634.91	417.15	51.35%	▲ 16.7	▲ 4.54%

Sub-Category	COGS Total	CPU (COGS per Unit)	CESR	CPU Δ	CESR Δ
Printers	\$8,504.00	472.44	42.86%	▼ 179.0	▲ 17.86%
Paper	\$8,408.25	467.13	40.00%	▲ 132.0	▲ 11.43%
Chairs	\$6,017.91	300.90	57.14%	▲ 10.1	▲ 7.14%
Accessories	\$5,654.05	353.38	75.00%	▼ 61.7	▲ 25.00%
Bookcases	\$4,616.59	577.07	33.33%	▲ 111.3	▼ 6.67%
Tables	\$3,522.46	391.38	75.00%	▼ 44.2	▲ 41.67%
Binders	\$3,438.93	429.87	100.00%	▲ 103.7	▲ 0.00%
Phones	\$3,259.06	543.18		▲ 87.3	▼ 60.00%
Storage	\$1,213.66	303.42		▲ 92.9	▼ 33.33%
Total	\$44,634.91	417.15	51.35%	▲ 16.7	▲ 4.54%

51.35%

CESR

▲ 4.54%

CESR by Region



\$44.63K

COGS Total

417.15

CPU (COGS per Unit)

417.15

CPU (COGS per Unit)

▲ 16.7

Segment	COGS Total	CPU (COGS per Unit)	CESR	CPU Δ	CESR Δ
Corporate	\$17,338.92	422.90	60.00%	▲ 74.0	▲ 22.50%
Home Office	\$15,138.68	409.15	46.15%	▲ 14.6	▼ 3.85%
Consumer	\$12,157.31	419.22	44.44%	▼ 43.4	▼ 9.40%
Total	\$44,634.91	417.15	51.35%	▲ 16.7	▲ 4.54%

CPU (COGS per Unit) by Region

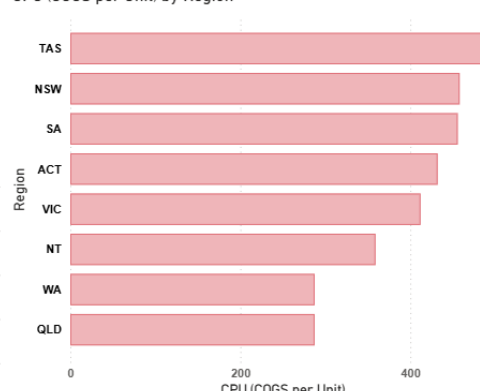


Figure 4 Internal Process Efficiency KPIs

The first KPI is *COGS per Unit (CPU)*. CPU measures the average cost we incur to procure and fulfil one unit. It links directly to Jubilee's strategic objective of protecting margins through disciplined buying and mix management: every dollar added to CPU erodes Gross Profit Margin (GPM) and, downstream, Operating Profit Margin (OPM). In Nov-2024, CPU was \$417.15 ($\Delta\text{MoM} = +16.7$), signalling cost pressure despite stable volumes. I visualised this with (i) a bar chart by Region to surface where costs are structurally high (TAS/NSW), and (ii) variance cards to emphasise month-on-month movement. Relevant fields include COGS and Units from Orders, with Date for time deltas, and Region/Segment/Category for slicing. CPU connects to other KPIs by explaining part of the variance in GPM (higher CPU to lower GPM) and by interacting with customer KPIs such as OTD (rush freight can lift CPU).

The second KPI is the *Cost-Efficiency Success Rate (CESR)*. CESR is the % of units purchased at or below their target CPU (negotiated price list). It operationalises procurement compliance and supplier performance, aligning to the strategy of scaling through repeatable, cost-controlled processes. In Nov-2024, CESR reached 51.35% ($\Delta\text{MoM} = +4.54$ pp): compliance improved, but not enough to offset CPU inflation. I presented CESR with a single KPI card and a column chart by Region to

compare adherence (Corporate leads; Home Office lags). Data required are a CPU Target table (by SKU/Category/Supplier), an At/Below Target flag per order line, and the same conformed dimensions as CPU. CESR links to supplier scorecards and predicts future CPU and GPM: higher CESR tends to pull CPU down and stabilise margins. Together, CPU and CESR give leaders a clear line of sight from daily buying behaviour to strategic profitability outcomes, while the dimensional cuts indicate precisely where to act.

3. Employee learning and growth

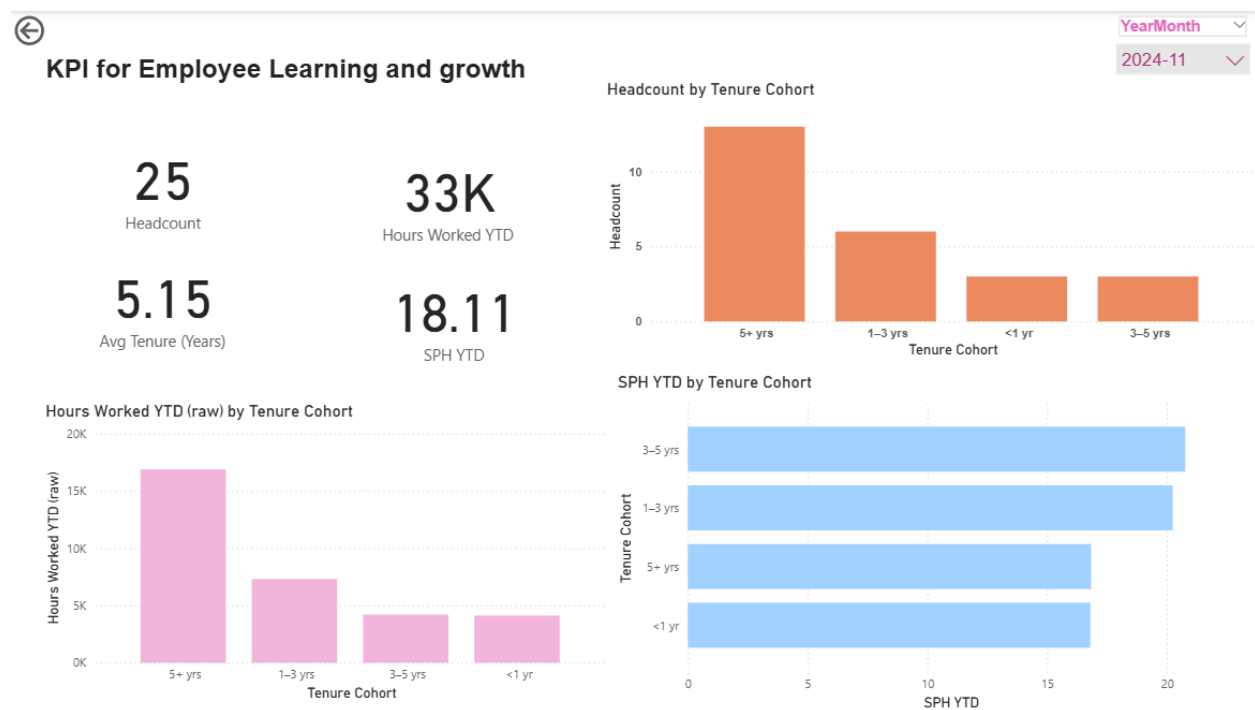


Figure 5 Employee Learning And Growth KPIs

The first KPI in this category is *Average Tenure (Years)*. It is calculated as the mean of *Salespersons[Years of Service]* and visualised with a headcount breakdown by tenure cohort (<1 yr, 1–3 yrs, 3–5 yrs, 5+ yrs). In Nov-2024, Average Tenure equals 5.15 years, with the largest share of staff in the 5+ years cohort. This KPI is selected because tenure is a leading indicator of organisational capability, knowledge retention, and service quality—key enablers of the firm’s strategic objectives of profitable growth and reliable fulfilment. Longer tenure generally reduces onboarding costs and error rates, and should support Operating Profit Margin (OPM) and On-Time

Delivery (OTD) on the finance and CSAT pages. Relevant fields include *Salespersons[Years of Service]* and the slice parameters (Region, Segment, Ship Mode, Category) to diagnose where experience is concentrated or thin.

KPI 2 - Sales per Hour, Year-to-Date (SPH YTD). This operational efficiency KPI captures the revenue generated per hour worked: $SPH\ YTD = \frac{SUM(Orders[Sales])}{SUM(Salespersons[Hours\ Worked\ YTD])}$. In Nov-2024 SPH YTD ≈ 18.11 , with the 1–3 yrs and 3–5 yrs cohorts outperforming on a per-hour basis, while 5+ yrs carry the highest Hours YTD but do not always lead on SPH. That pattern suggests two actions: (1) rebalance opportunities so high-SPH cohorts receive sufficient pipeline; (2) codify mentoring to steepen the learning curve for the <1 yr cohort. Slice SPH by Region/Segment/Category and validate that higher SPH translates into better OPM (not just higher sales). The company need fields: *Orders[Sales]*, *Salespersons[Hours Worked YTD]*, *Salespersons[Years of Service]*, joined via *Salesperson ID*.

4. Social performance

KPI for Social Performance

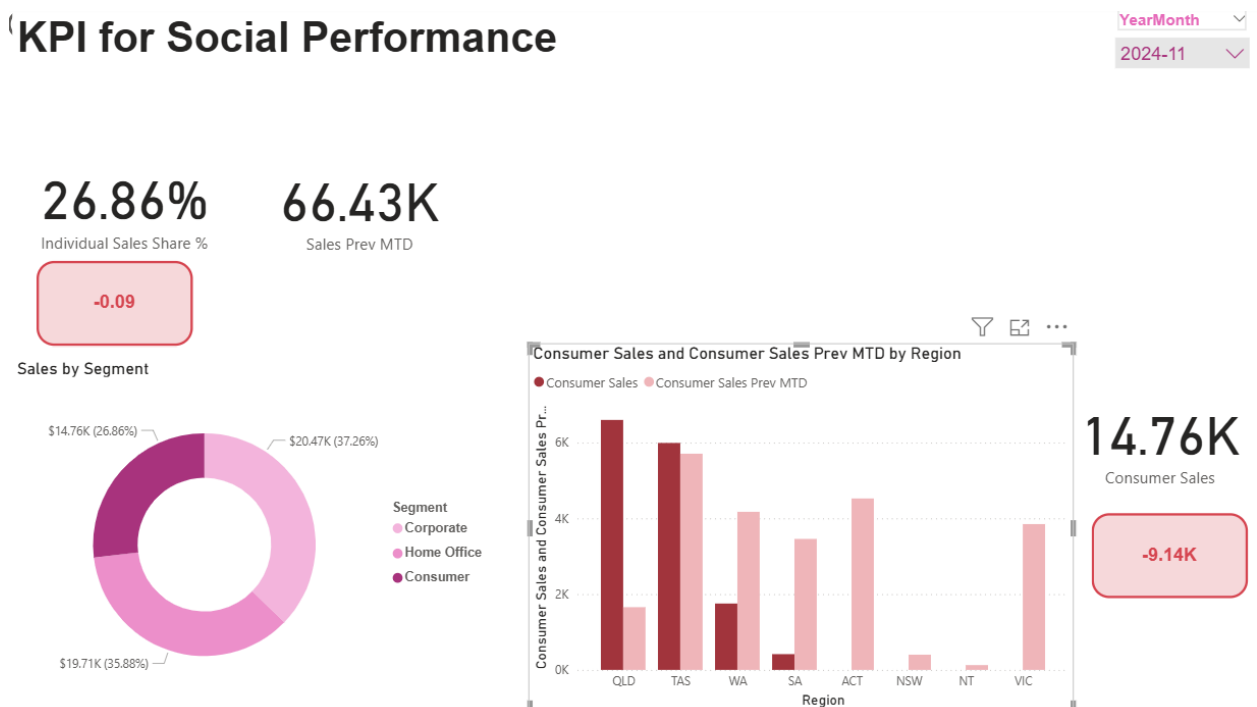


Figure 6 Social Performance KPIs

Our social performance view uses two complementary KPIs built directly from the model to show how well Jubilee reaches individual shoppers. The first KPI is

Individual Sales Share (ISS), defined as Consumer sales divided by total sales; it ties to the strategic goal of broad community access by quantifying the revenue mix coming from individual customers rather than Corporate or Home-Office accounts. In Nov-2024 the ISS is 26.86%, and the month-over-month card indicates a decline of roughly nine percentage points, signalling a shift of revenue back toward business buyers. The donut confirms that Home Office (~37%) and Corporate (~36%) dominate the mix, while Consumer is only ~27%.

The second KPI, Consumer Sales with MoM change, tracks the size and momentum of that retail stream; Consumer Sales total \$14.76K with a -\$9.14K MoM drop, and the regional chart shows QLD and TAS holding up best while ACT, SA and WA lag. Both KPIs are calculated with standard fields already in the model - sales amounts in the Orders table, a Customer Segment attribute to identify "Consumer", the Date table to derive the previous month, and Region for the breakdown so they are reproducible and auditable. Taken together, the numbers imply that the company is under-delivering on its inclusion objective this month; a pragmatic response is to defend QLD/TAS with basket-builder bundles and free-shipping thresholds, run re-activation offers to lapsed Consumer shoppers in ACT/SA/WA, and rebalance merchandising toward Consumer-friendly categories now skewed to business buyers, then monitor weekly ISS and Consumer-Sales-MoM to confirm recovery.

5. Environmental sustainability

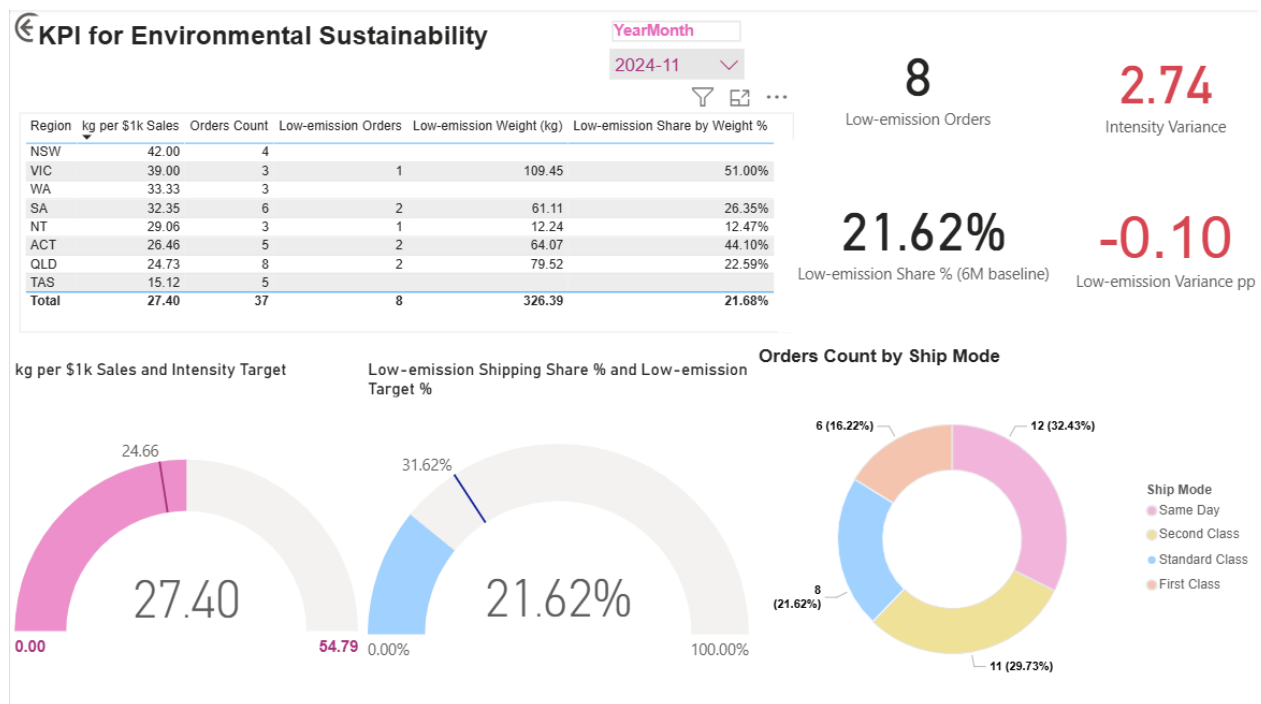


Figure 7 Environmental Sustainability KPIs

KPI 1 – Shipping Weight Intensity (kg per USD 1,000 sales).

This KPI tracks the environmental “intensity” of fulfilment: total shipped kilograms per USD 1,000 of revenue. Lower values imply leaner packaging and a product mix that places less weight into the transport network, which typically reduces logistics emissions. We compute it as the sum of Quantity × Weight(kg) at line level, divided by (Sales/1,000). Visuals include a gauge (Actual vs Target), a regional breakdown, and a monthly trend. The Target is set dynamically as the rolling 6-month baseline × 90% (a 10% reduction goal).

Nov-2024 result: 27.40 kg/1k\$, with Intensity Variance +2.74, indicating performance above (worse than) target. Priority actions: optimise packaging in high-intensity categories/regions and review the use of expedited modes that increase total shipped weight.

KPI 2 – Low-emission Shipping Share (% Standard Class).

This KPI monitors the proportion of Standard-Class shipments as a proxy for lower-emission ground modes versus expedited/air. It is calculated as Standard-class orders ÷ total orders (or, optionally, by shipped weight). Visuals include a card, a gauge (Actual vs

Target), a ship-mode donut, and a regional table. The Target is set dynamically as (rolling 6-month baseline) + 10 percentage points.

Nov-2024 result: 21.62% vs Target 31.62% (–10 pp), with most of the shortfall concentrated in a few regions. Actions: default to Standard for eligible SKUs/regions, surface expected delivery dates to reduce unnecessary upgrades, and pilot product bundling to cut weight per order.

These KPIs operationalise Jubilee’s environmental pillar in the Balanced Scorecard: reducing logistics footprint while protecting profitability. They translate the sustainability plan into trackable targets and concrete levers (packaging optimisation, shipping-mode policy) that can be owned by Supply Chain and Store Operations and reviewed monthly alongside financial KPIs.

6. Indigenous data sovereignty principles and ethical data collection and reporting practices

Indigenous Data Sovereignty KPIs.

KPI 1 – CARE/FPIC Governance Compliance Rate: Percentage of dashboards that use Indigenous-identifying fields and fully comply with the CARE Principles and FPIC/consent under the AIATSIS Code (Kukutai & Taylor, 2016; Williamson et al., 2023). Data: a governance checklist (ReportID, use of Indigenous data, consent, CARE C/A/R/E, safe reporting, incident, reviewer/date). Formula: compliant items ÷ items using Indigenous-identifying data. Target: 100%.

KPI 2 – Culturally-Safe Reporting Coverage (and Incident Rate): Share of dashboards that present Indigenous data only in aggregated, de-identified form with a <5 cell-suppression rule and appropriate access controls; Incident Rate counts any breach/complaint (Reconciliation Australia, 2021; United Nations, 2013). Formula: safe items ÷ items using Indigenous-identifying data; Target: 100% safe, 0 incidents.

Current status: N/A – no dashboards currently use Indigenous-identifying data. These KPIs translate our RAP commitments into auditable controls and will be reviewed monthly by the Data Steward and RAP Working Group.

[Words Count: 1560]

Reference

- Kukutai, T., & Taylor, J. (Eds.). (2016). *Indigenous data sovereignty: Toward an agenda*. ANU Press. <https://doi.org/10.22459/CAEPR38.11.2016>
- Williamson, B., Provost, S., & Price, C. (2023). Operationalising Indigenous data sovereignty in environmental research and governance. *Environment and Planning F: Philosophy, Theory, Models, Methods and Practice*, 2(1–2), 281–304. <https://doi.org/10.1177/26349825221125496>
- Reconciliation Australia. (2019, November 21). *How RAPs advance reconciliation* [Video]. YouTube.
- Reconciliation Australia. (2021, January 15). *State of reconciliation in Australia: Moving from safe to brave*. Reconciliation Australia.
- United Nations. (2013). *The business reference guide to the UN Declaration on the Rights of Indigenous Peoples*. United Nations Global Compact.

Appendix — Governance Checklist

ReportID	Uses_Indigenous_Data	FPIC_Consent	CARE_C	CARE_A	CARE_R	CARE_E	Safe_Reporting	Incident
Customer Satisfaction	False	False	False	False	False	False	False	False
KPI for Environmental Sustainability	False	False	False	False	False	False	False	False

IDS_Governance_Log is a small metadata table (one row per dashboard/page) used to compute two governance KPIs for Indigenous data sovereignty. It records whether a page uses Indigenous-identifying data (Uses_Indigenous_Data), has FPIC/consent, meets CARE requirements (CARE_C/A/R/E), applies culturally-safe reporting (Safe_Reporting - aggregated, de-identified, <5 cell suppression, access controls), and any Incident. From this log: CARE/FPIC Governance Compliance % = compliant rows ÷ rows using Indigenous-identifying data; Culturally-Safe Reporting Coverage % = safe-reporting rows ÷ rows using Indigenous-identifying data; Incident Rate = count of incidents. Targets: 100%, 100%, and 0 respectively. In this submission all rows are *FALSE*, so KPIs display N/A; when a page introduces an Indigenous breakdown, populate the row and the KPIs calculate automatically. The checklist and KPIs are grounded in the CARE Principles and the AIATSIS Code of Ethics (self-determination,

cultural safety), align with UNDRIP/FPIC guidance, and operationalise the organisation's Reconciliation Action Plan (RAP) commitments (Kukutai & Taylor, 2016; Williamson, Provost & Price, 2023; Reconciliation Australia, 2021; United Nations, 2013).